

Report Lab 1 : Database

Adam Naji, Souhayl Moujahid, Ahmad Khriss, Ilyas Mouhri

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1 Role

Adam Naji	group manager (worked on : Reseach Project, Calibration records and Maintenance Operation diagrams)
Ahmad Khriss	group integrator (worked on : Laboratory and Equipment diagrams)
Souhayl Moujahid	diagram designer (worked on : People, Reservations and Budgets diagrams)
Ilyas Mouhri	diagram designer (worked on : Certifications and Consumables diagrams)

2 Project Challenges

2.1 Difficulty 1: Team work

The first challenge we faced is how to organize teamwork. It is our first time in the College of Computing, where we had to work in teams.

To face this challenge, we created a WhatsApp channel for better communication, divided the work between us according to our skills, and agreed on one person to fuse the work.

2.2 Difficulty 2: Synchronization

The second challenge of this lab was to find the perfect solution for creating and collaborating on our work because even though everyone on the group has to do his own part, we must synchronize with what was done by our teammate.

In order to answer this problem, we decided to work on Google Slides, an auto-synchronized presentation builder online. Thanks to it, we could see what others were building and our group integrator could easily merge everyone's sub-diagrams

3 Justification of modeling choices

We used ISA hierarchies to represent the specializations described in PE2 and PE4, so that student, employee, academic, non-academic, administrative and technical staff inherit the common attributes of People while keeping their own. We modeled Calibration record and Maintenance Operation as weak entities

because CA4 and MA2 state that they are identified only in combination with their equipment unit, using Calibration date and maintenance start as partial keys. We separated Equip Model from Equip Units as required by EQ1 and EQ5,

and we used an aggregation over the paired relationship between consumables and Laboratories because CO8 assigns a technical staff member to a laboratory-consumable pair rather than to a consumable alone.

4 Assumptions

We assumed that the specialization of People into student and employee is total and disjoint, since PE2 states that a person is either one or the other. We assumed that only the current state is stored, following PE5, EQ4 and CE5, so no history of reporting lines, equipment moves or past certifications is kept. We also assumed that a person does not have to be attached to a laboratory or to take part in a project, because LA3 and PR2 only constrain the laboratory and the project side.

5 Requirements that could not be expressed in the ER model

Several constraints restrict which subtype may take part in a relationship, and the ER model cannot express this: PR2 and RE1 exclude administrative staff, BD2 allows only academic employees to manage budgets, and PE5 states that the dean reports to no one while all faculty and administrative staff report to the dean. Counting and cross-relationship constraints are also out of reach, such as PR3 (each project must have exactly one PI and at most one Co-PI), CE7 (only people holding the required certification may use an equipment model) and CO10 (a consumable is acquired only within an existing reservation). Finally, the ER model cannot enforce attribute domains, so the enumerated values in PR1, EQ3, RE2, MA2 and CA2 must be checked by the application.

6 Final Result

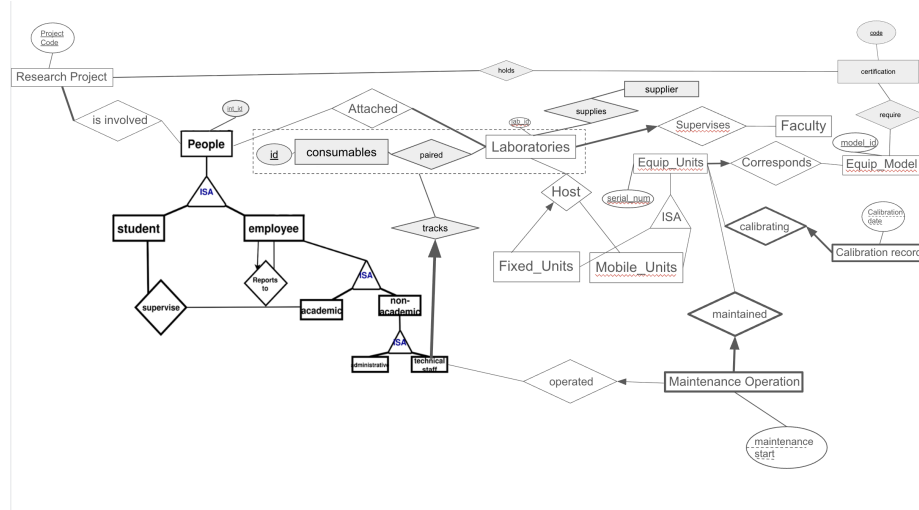


Figure 1: Complete Diagram